3 **(a)** Two light identical springs are joined in series and a 120 g bar magnet is hung from the lower end as shown in Fig. 3.1. The effective spring constant of the combined springs is 11 N m^{-1}.

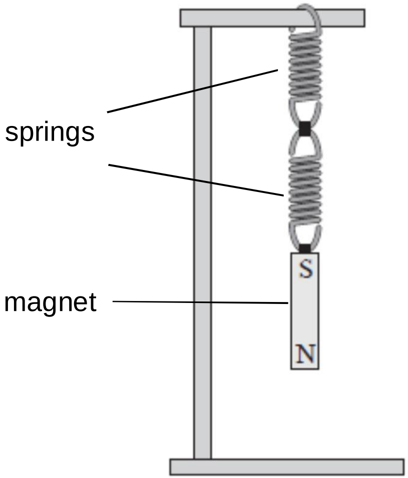

Fig. 3.1

The bar magnet is displaced a small distance vertically downward from its equilibrium position and released. It then oscillates in simple harmonic motion.

(i) By considering the forces on the magnet at its equilibrium position, and also at a small displacement x downward from its equilibrium position, show that the oscillation is a simple harmonic motion where the acceleration a is given by the equation

$$a = -92\ x$$

[4]

(ii) Hence calculate the frequency at which the magnet oscillates.

frequency = ______________________________ Hz [2]

(b) Two systems in part **(a)** are now suspended with the North pole of their magnets located inside a coil of wire. The two coils are connected together with conducting leads as shown in Fig. 3.2.

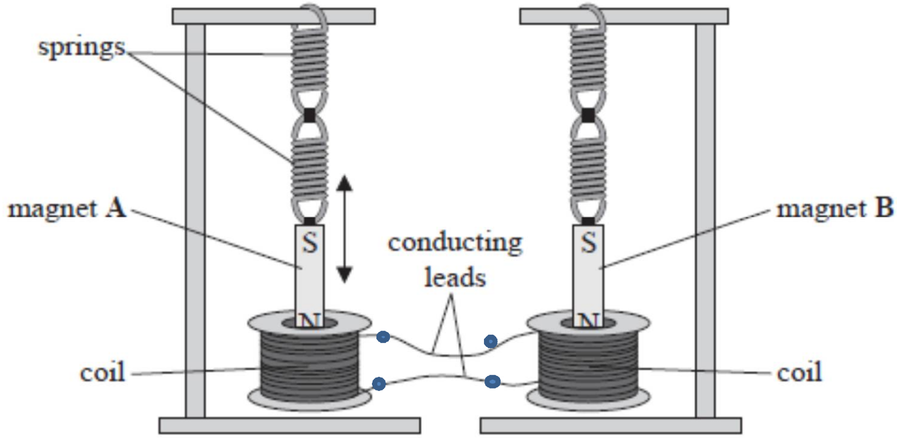

Fig. 3.2

Magnet A is displaced so that it oscillates vertically. The North pole of magnet A moves into and out of the coil of wire with simple harmonic motion. As this motion continues, magnet B starts to oscillate. The amplitude of oscillation of magnet B increases over time.

Explain why magnet B starts to oscillate with an increasing amplitude.

--- [5]